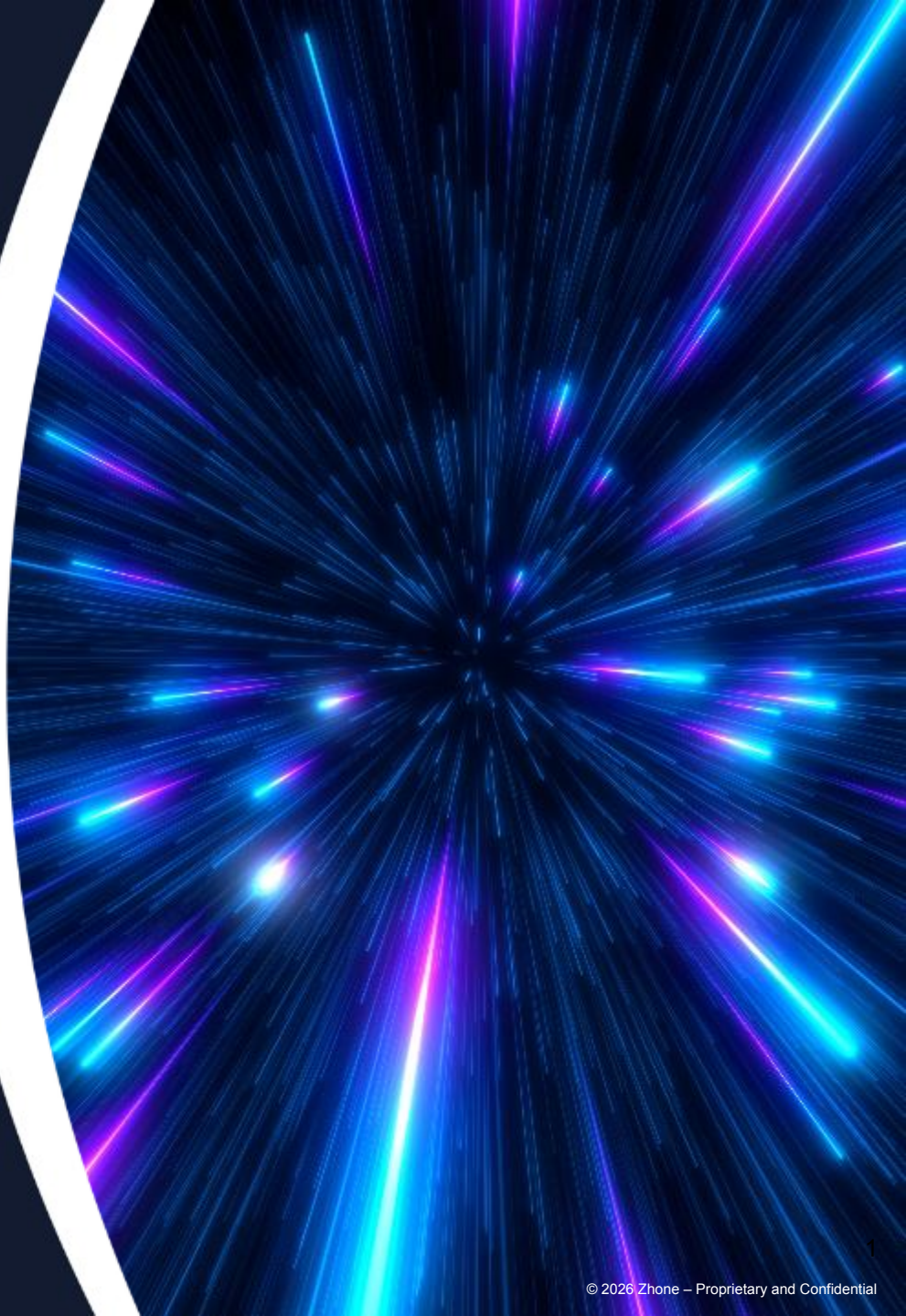
Network Devices, MAC & ARP

Presenter: Nguyen Manh Hung
Date: 06/11/2026

Embedded Trainees



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OVERVIEW

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1. hub
2. bridge
3. switch
4. router
5. firewall

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MAC Address

1. MAC Denifition, Purpose
2. MAC Types
3. MAC Address Table

3

ARP Protocol

1. ARP Definition
2. How ARP works
3. ARP Packet Structure
4. ARP Table

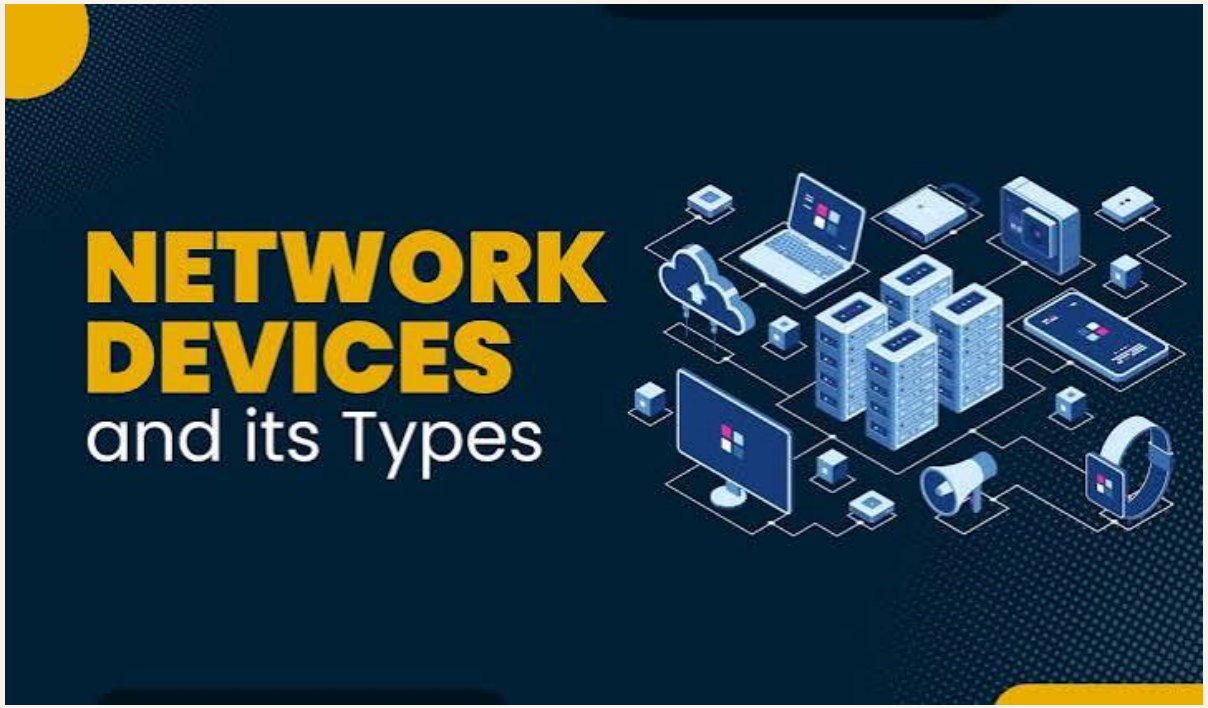
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01

HARDWARE

1. Network devices

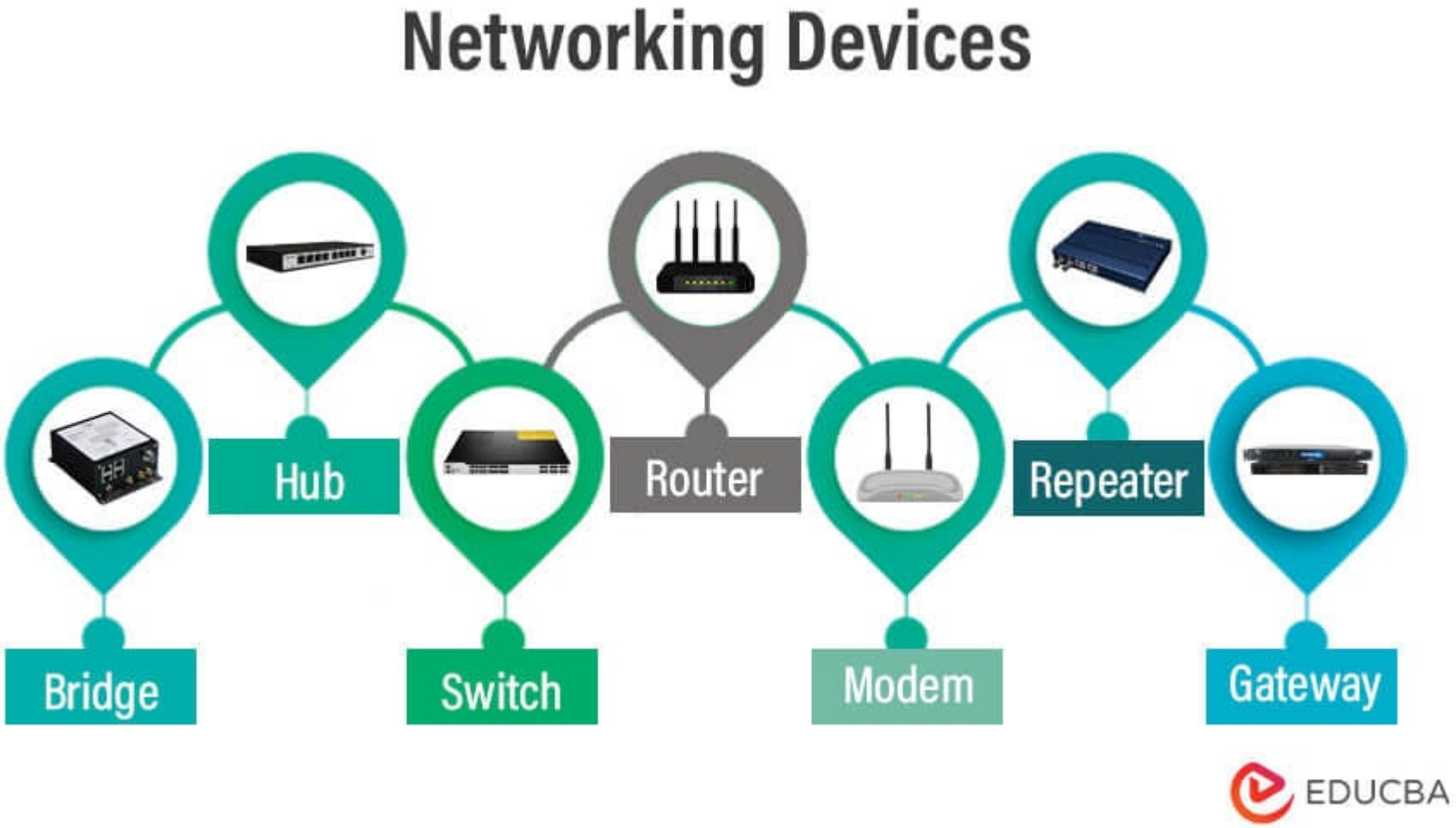
Physical infrastructure provides the foundation for all network traffic.



https://www.networkacademy.io/ccna/network-fundamentals/hubs-bridges-switches

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1. Network devices



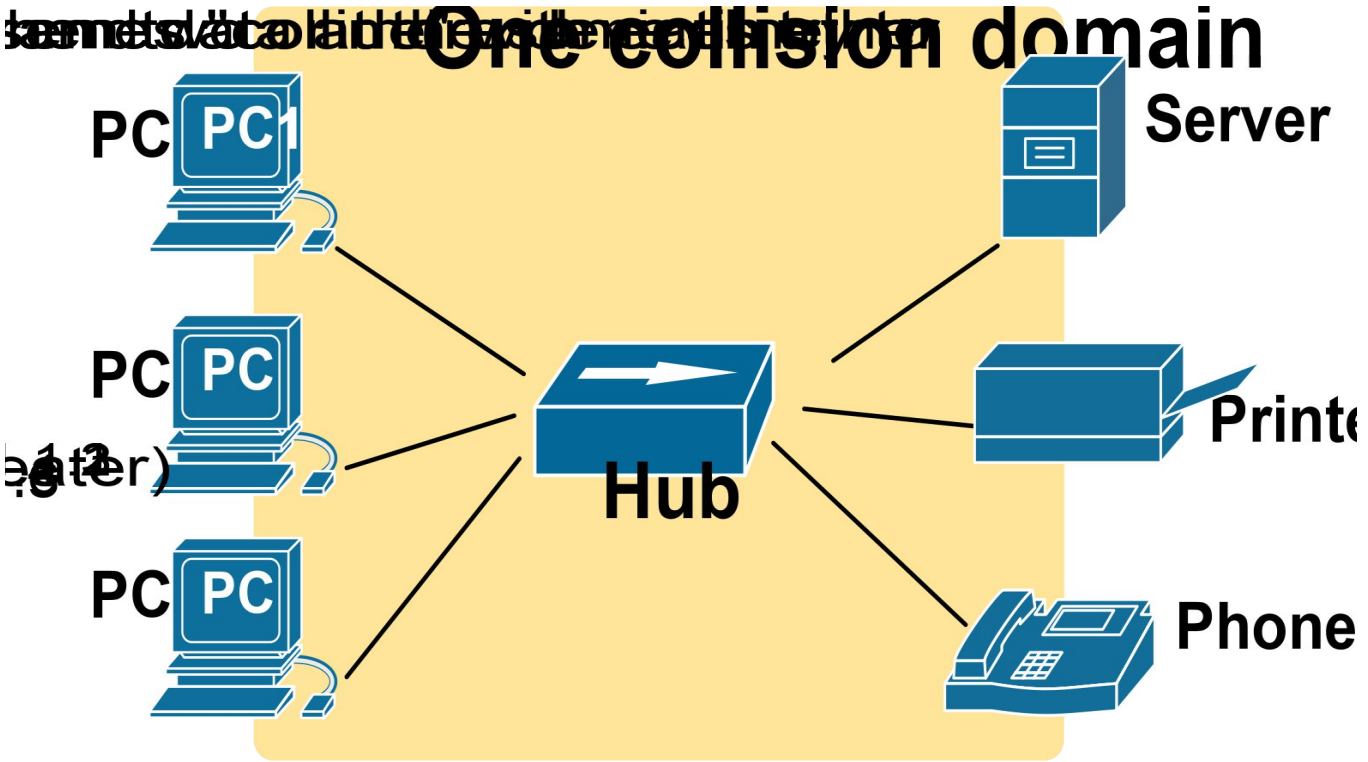
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1. Network devices

Hub

Hub is a device that connects multiple devices and computers together. Hubs can also be called repeaters or concentrators because they act as the center of a local area network (LAN).

- Layer 1
- It takes an incoming signal on one port and broadcasts it to all other ports without knowing the recipient.
- High network traffic collisions and security risks (everyone sees everyone's data).
- It is a obsolete device, replaced by ***Switches***.

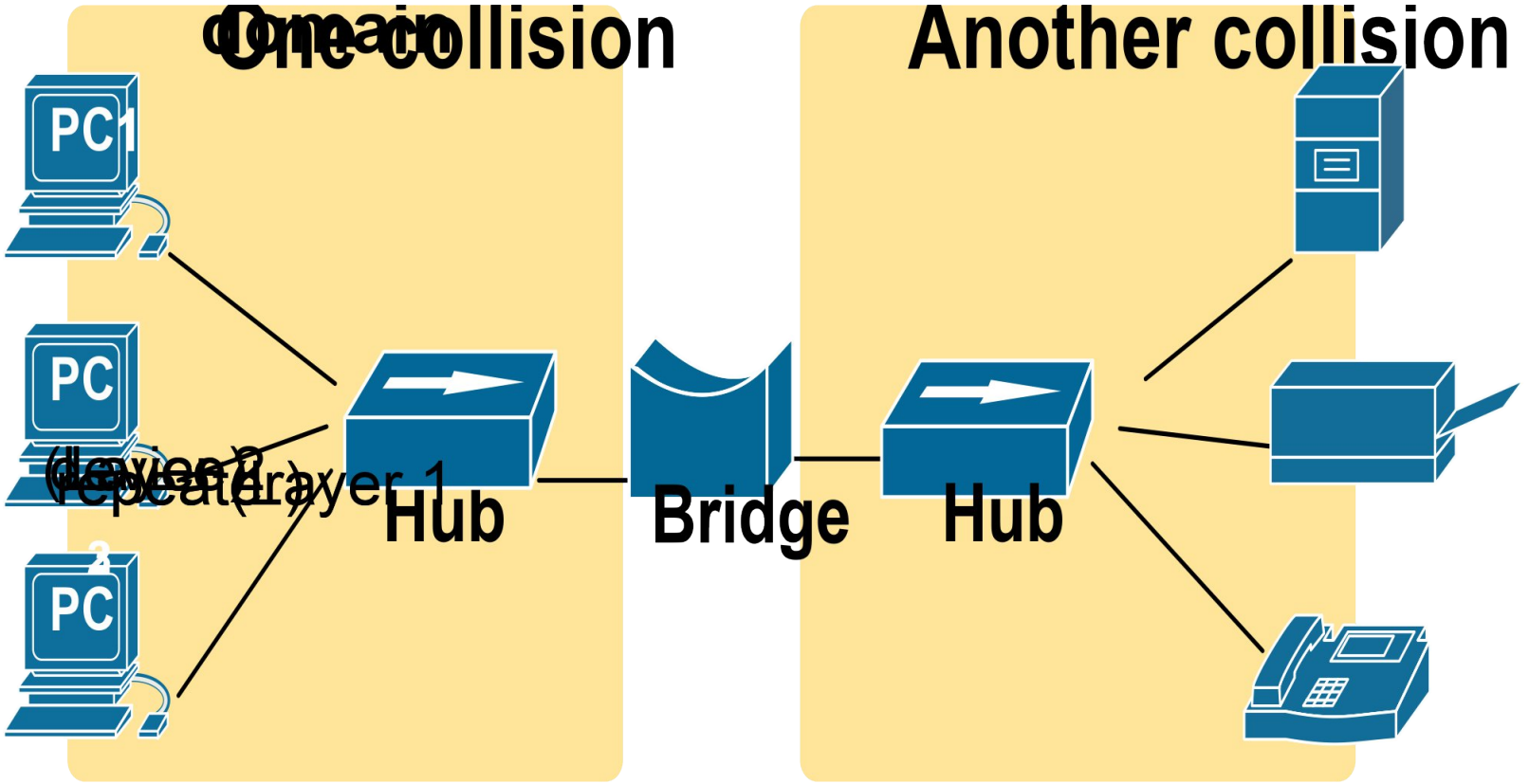


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1. Network devices

Bridge

A bridge is a networking device that connects two or more Local Area Network (LAN) segments and enables communication between them.



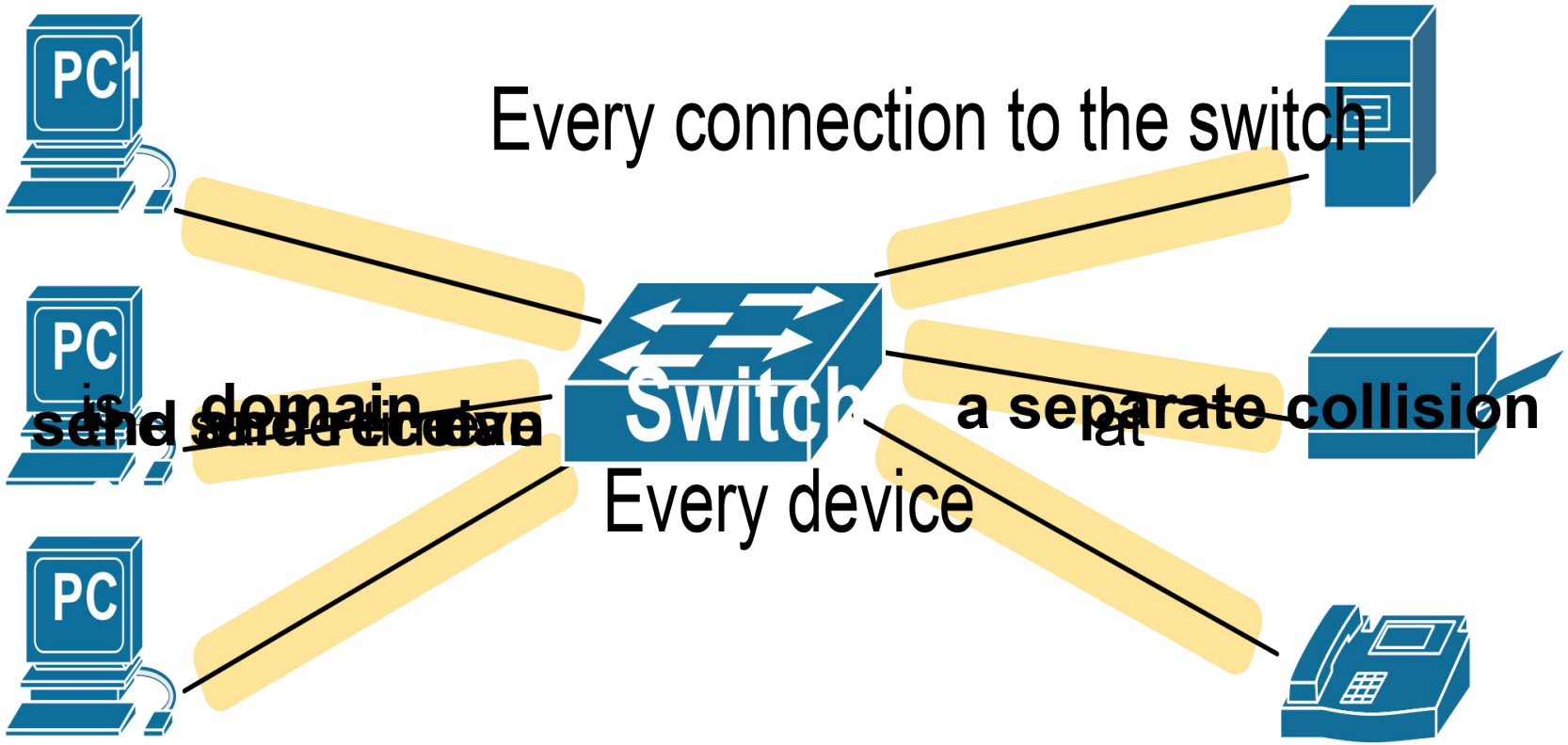
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1. Network devices

Switch

A Network Switch is an intelligent networking device that main function is to connect multiple devices (such as computers, printers, servers, PLCs) within the same LAN.

- Layer 2
- Unlike ***hub***, switch learns the MAC address of every connected device.
- It sends data only to the specific port where the destination device is connected.
- Reduces collisions and improves security and speed.

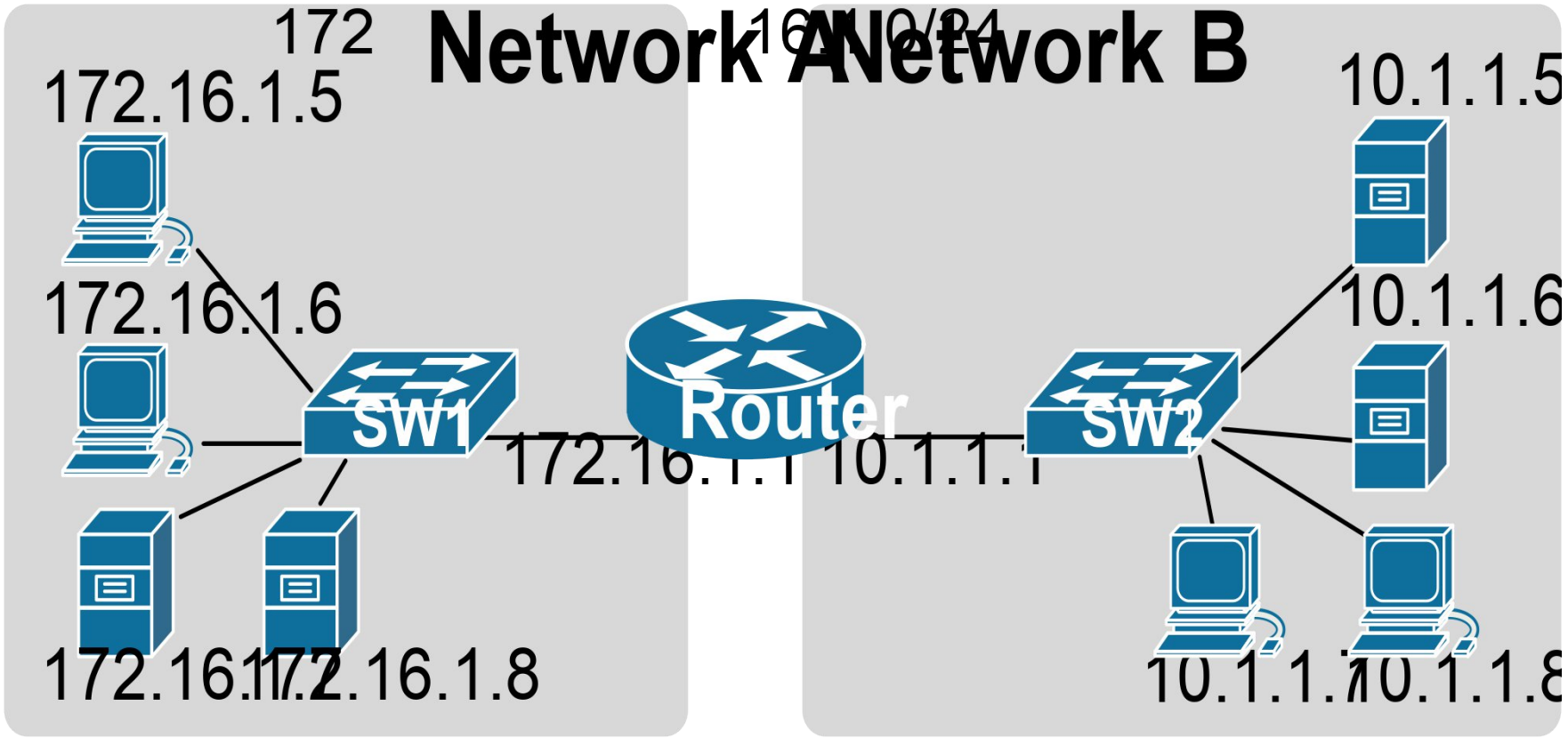


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1. Network devices

Router

A router is a networking device that forwards data packets between different computer networks. It allows multiple devices to share an Internet connection efficiently.

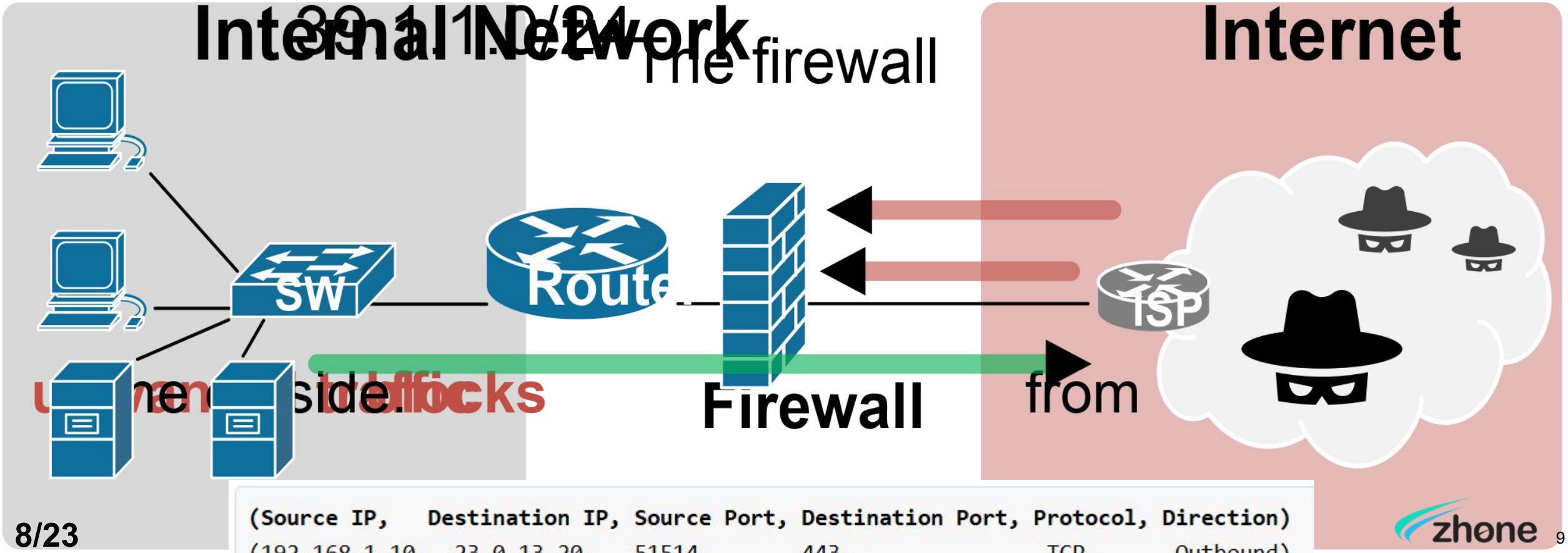


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1. Network devices

Firewall

It is a network security system, comprising hardware, software, or both that monitors and controls incoming and outgoing network traffic based on predetermined security rules.



(Source IP,	Destination IP,	Source Port,	Destination Port,	Protocol,	Direction)
(192.168.1.10,	23.0.13.20,	51514,	443,	TCP,	Outbound)

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1. Network devices

Compare Hub, Switch and Router

Feature	Hub	Switch	Router
Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	None (Bits)	MAC Address	IP Address
Data Flow	Broadcast (to all)	Unicast (to destination)	Route (best path)
Intelligence	Dumb	Smart	Smarter
Used For	Obsolete	Connecting Devices (LAN)	Connecting Networks (WAN)

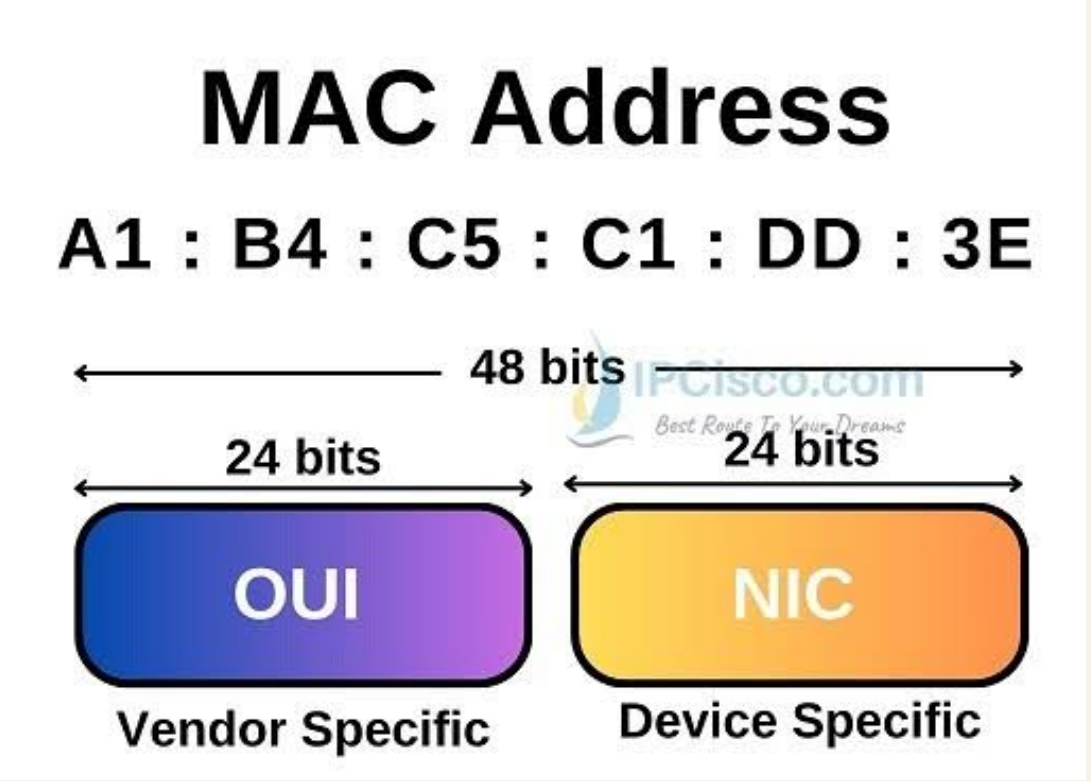
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02

ADDRESSING

2. MAC Address

Physical addresses ensure hardware-specific identification on the local network



2.1. MAC Denifition

2.2. What is the purpose of MAC address?

2.3. MAC Types

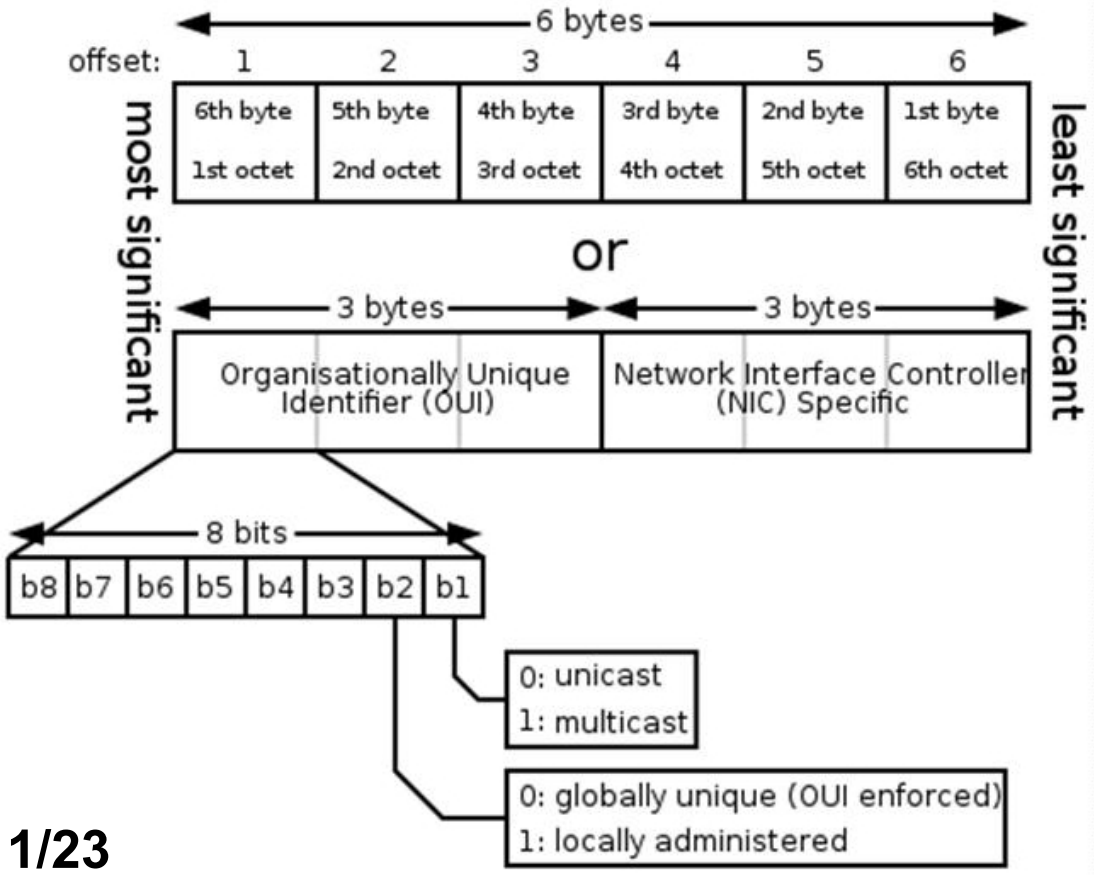
2.4. MAC Address Table

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2. MAC Address

2.1 What is MAC address?

A MAC address (Media Access Control address) is a physical or hardware address assigned to each network device.



- A MAC address is a 48-bit (6-byte) binary string, written as 12 hex characters in pairs, separated by hyphens, colons, or periods (00:1A:2B:3C:4D:5E).
- **OUI**: Organizationally Unique Identifier; first 24 bits. The vendor ID issued by IEEE.
- **NIC-Assigned**: Vendor-Assigned last 24 bits. Set by the vendor for each physical card; one OUI supports up to 2^24 devices.

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2. MAC Address

2.2 Why is a MAC address needed?

- Device identification in a local area network (LAN) : MAC addresses enable network switches to determine exactly which piece of hardware should receive a given data packet, ensuring there is no confusion.

- MAC Filtering: A network administrator or a Wi-Fi modem can be configured to allow only those devices with MAC addresses included in a trusted list to connect to the network.

- Combined with an IP address to send packets from the source machine to the destination machine:

- Layer 2 uses MAC addresses and is responsible for packet delivery from hop to hop.
- Layer 3 uses IP addresses and is responsible for packet delivery from end to end.

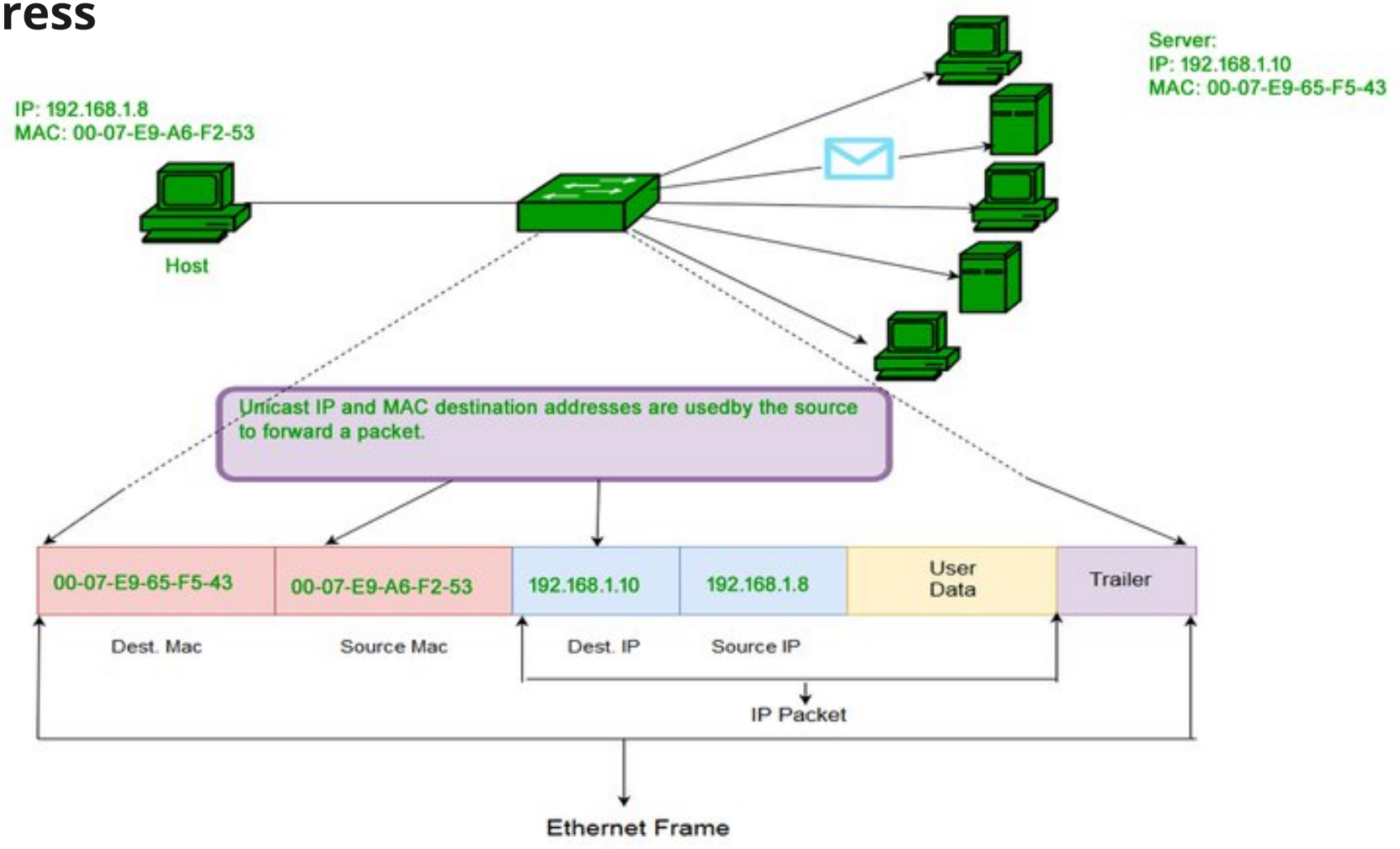
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2. MAC Address

2.3 MAC Types

Unicast MAC address

A4:5E:60:11:22:33



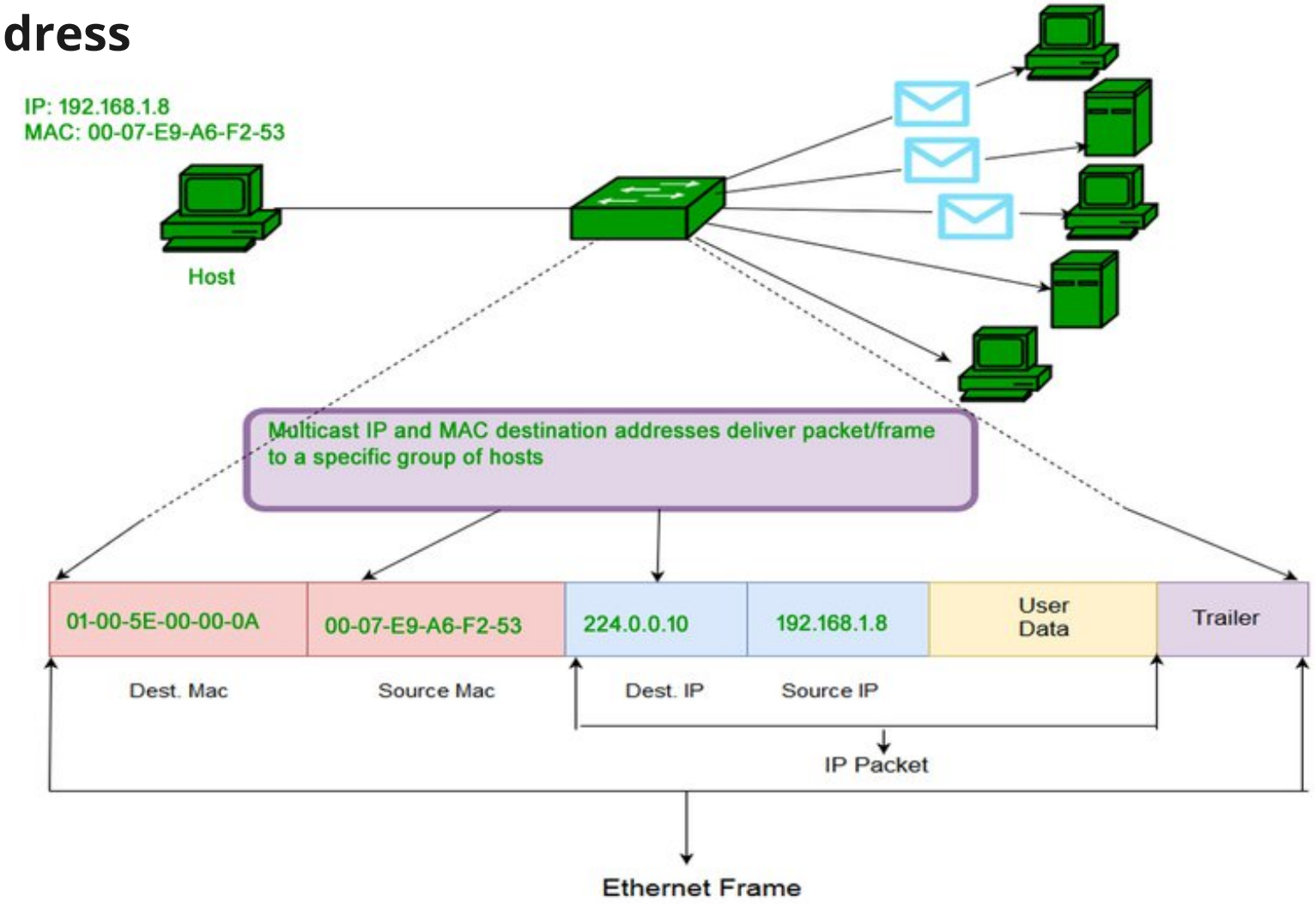
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2. MAC Address

2.3 MAC Types

Multicast MAC address

01:00:5E:00:00:0A



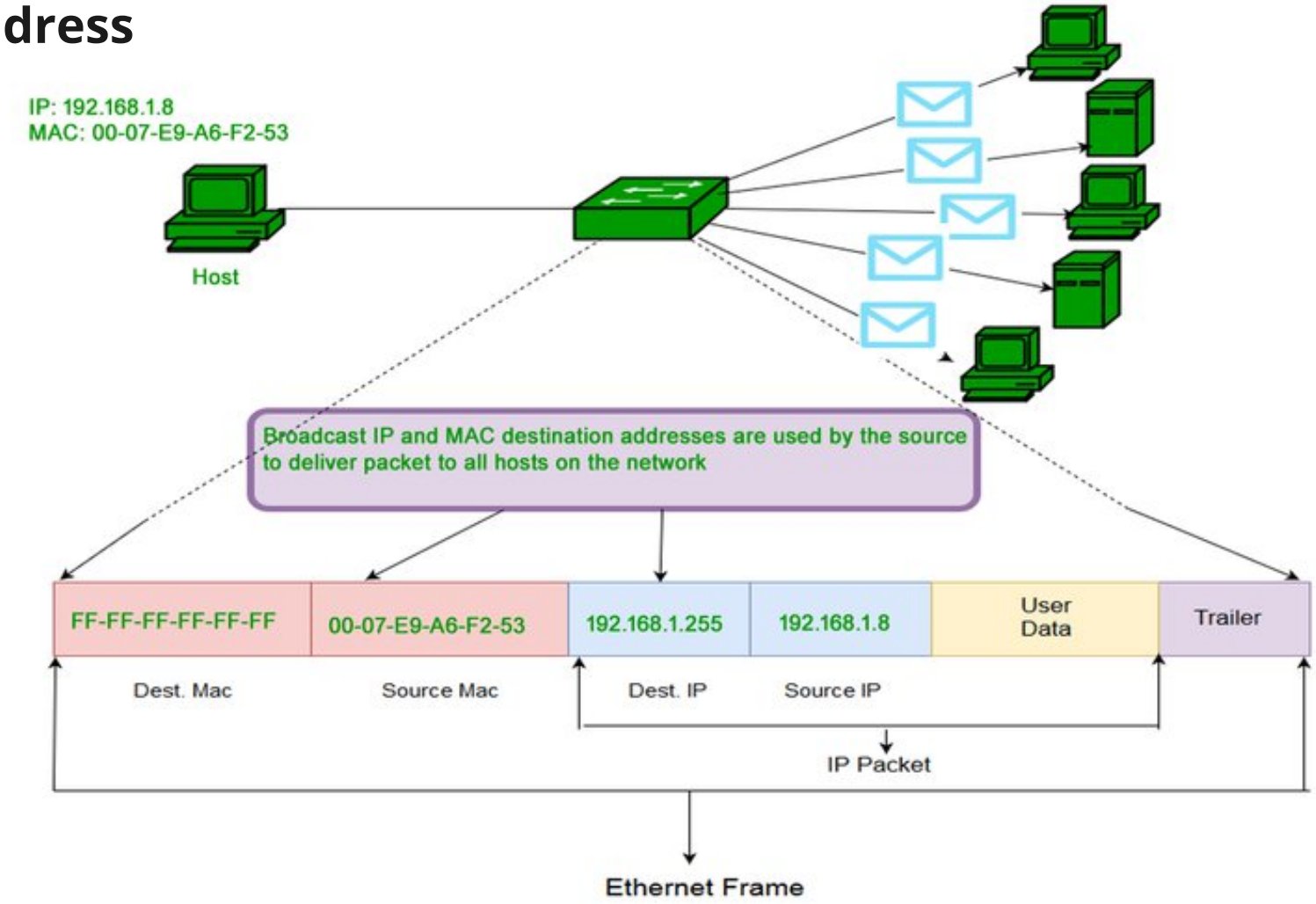
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2. MAC Address

2.3 MAC Types

Broadcast MAC address

FF:FF:FF:FF:FF:FF



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2. MAC Address

2.4 MAC Address Table

The MAC Address Table is a data table stored in the CAM (Content Addressable Memory) of a switch.

The table contains mappings between:

- Device MAC address
- Corresponding VLAN
- Connected port
- Entry type (dynamic or static)

```
SwitchX#show mac address-table
          Mac Address Table
------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
 All    0008.a445.9b40    STATIC      CPU
 All    0100.0ccc.cccc    STATIC      CPU
 All    0100.0ccc.cccd    STATIC      CPU
 All    0100.0cdd.dddd    STATIC      CPU
   1    0008.e3e8.0440    DYNAMIC     Fa0/2
Total Mac Addresses for this criterion: 5
SwitchX#
```

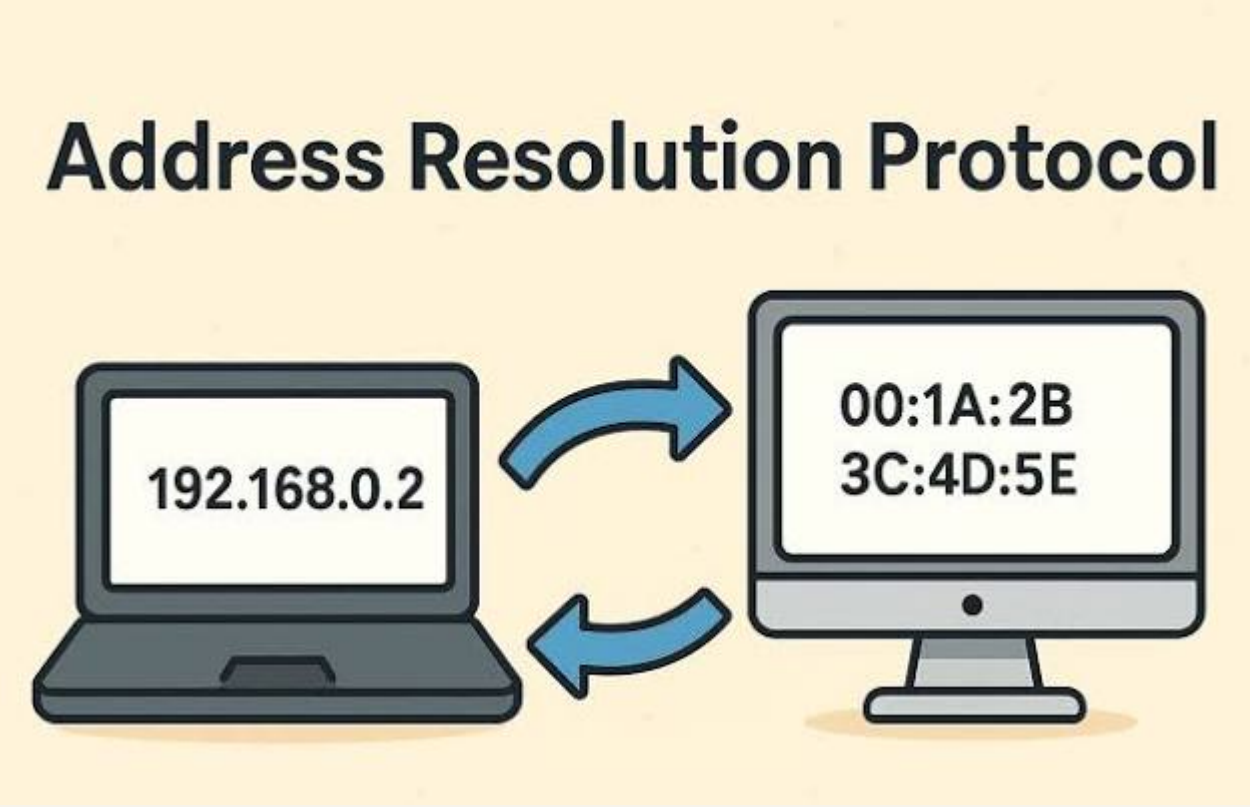

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03

PROTOCOLS

3. ARP Protocol

ARP is the translation layer required to bridge IP logical addresses to hardware physical addresses.



3.1 ARP Definition

3.2 ARP Packet Structure

3.3 How ARP works

3.4 ARP Table

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3. ARP Protocol

3.1 ARP Definition

ARP (Address Resolution Protocol) is a network protocol used to map a network address (an IPv4 address) to a physical address (a MAC address) within a local area network (LAN).

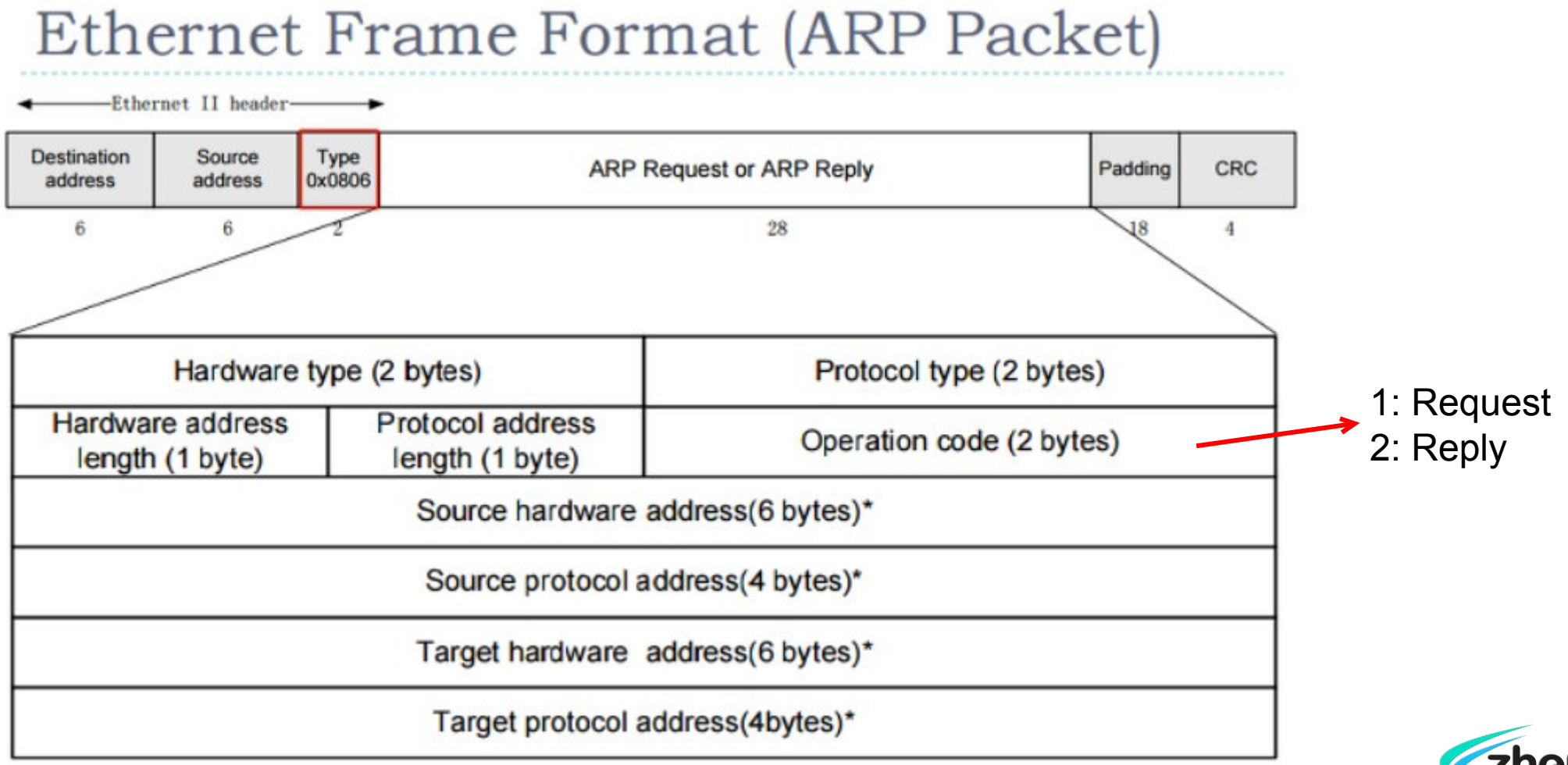
The ARP protocol consists of four main types:

- **Proxy ARP**: Designed for devices within a network, it operates within a local scope.
- **Gratuitous ARP**: This type of request helps identify duplicate IP addresses. Gratuitous ARP does not perform a standard IP-to-MAC address resolution request.
- **Reverse ARP**: Used by client systems within a LAN to request an IPv4 address from a router's ARP table.
- **Inverse ARP**: Commonly used in Frame Relay or ATM networks, Inverse ARP is employed to determine the IP address of a node based on its data-link layer address. It obtains the Layer 2 virtual circuit address through Layer 2 signaling.

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3. ARP Protocol

3.2 ARP Packet Structure

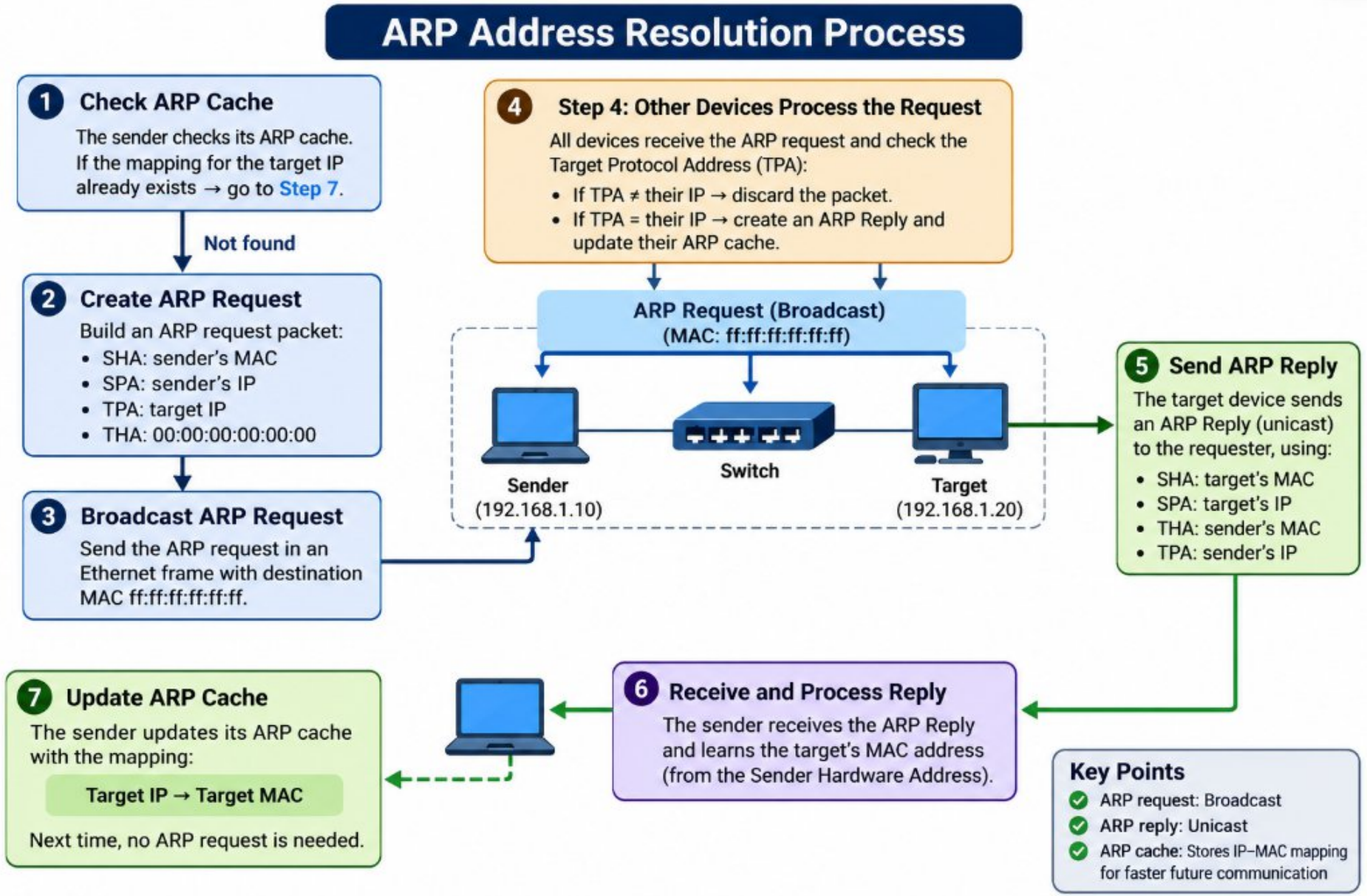


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3. ARP Protocol

https://github.com/hocchudong/thuctap012017/blob/master/TamNT/T%C3%ACm%20hi%E1%BB%83u%20giao%20th%E1%BB%A9c%20ARP.md

3.3 How ARP works



3. ARP Protocol

3.4 ARP Table

The ARP table is a table that stores mappings between IPv4 addresses and MAC addresses for devices with which a computer needs to communicate on a directly connected network.

The ARP table helps to:

- Reduce the number of ARP requests.
- Decrease network overhead.
- Increase data transmission efficiency.
- Store information about neighbors that have already been resolved.

router-1# **show arp**

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.3	-	0005.5EE0.8BA0	ARPA	Ethernet0/0
Internet	192.168.2.254	-	0001.C712.8A37	ARPA	Ethernet3/0
Internet	192.168.3.1	-	00D0.58C2.8ACB	ARPA	Ethernet1/0
Internet	192.168.3.2	11	0060.2FE9.4B11	ARPA	Ethernet1/0
Internet	192.168.3.3	11	0060.2F74.6770	ARPA	Ethernet1/0

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3. ARP Protocol

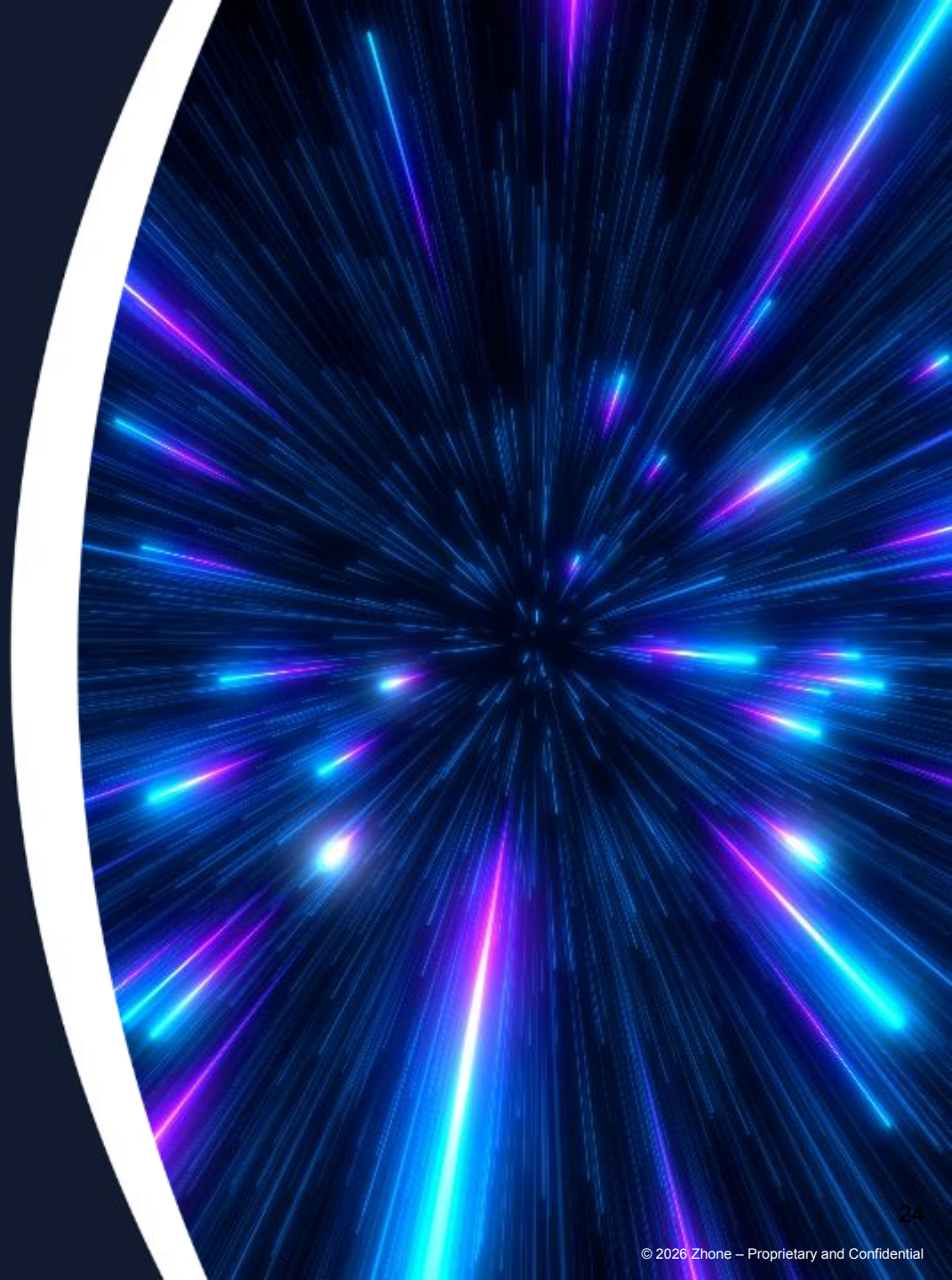
ARP Table vs Switch MAC Address Table

Feature	MAC Address Table (CAM Table)	ARP Table
OSI Layer	Layer 2 (Data Link Layer)	Layer 3 (Network Layer)
Primary Device	Used mainly by Switches	Used by Routers, PCs, and Servers
Core Mapping	Maps MAC Address to Physical Port	Maps IP Address to MAC Address
Main Purpose	Determines which port to forward a frame to reach the destination device.	Discovers the hardware address (MAC) of a device when only its software address (IP) is known.

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